

CSA0311 - DATA STRUCTURES

OUTPUT PREDICTION

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2nd Question

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The question list on the left shows 'Q2 Array' as the current question. The 'Your Code' area contains a C program for a binary search algorithm. The 'Input' field contains the values 3, 12 14 16, and 14. The 'Output' field shows the program's execution results.

Q2
Array
1 mark

Q3
Merging
1 mark

2/3 answered

Your Code

```
#include<stdio.h>
2 int main() {
3     int arr[100], n, i, key;
4     int front = 0, end, middle, found = 0;
5     printf("Enter the size : \n");
6     scanf("%d", &n);
7     printf("Enter the sorted elements: \n");
8     for(int i=0; i<n; i++) {
9         scanf("%d", &arr[i]);
10    }
11    printf("Enter the key: \n");
12    scanf("%d", &key);
13    end=n-1;
14    while(front<end) {
15        middle=(front+end)/2;
16        if(arr[middle]==key) {
17            found = 1;
18            printf("Element found at %d\n", middle + 1);
19            break;
20        }
21        else if (key<arr[middle]) end = middle - 1;
22        else end = middle + 1;
23    }
24    if(found==0)
25        printf("element not found");
26    return 0;
27 }
28 }
```

Submitted

Input

```
3
12 14 16
14
```

Output

```
Enter the size :
enter the sorted elements:
enter the key:
Element found at 2
```

3rd Question:

The screenshot shows the SIMATS Online Programming Lab interface for the 3rd question. The question list on the left shows 'Q3 Merging' as the current question. The 'Your Code' area contains a C program for merging two arrays. The 'Input' field contains the values 3, 1 2 3, 4, and 4 5 6 7. The 'Output' field shows the program's execution results.

Q2
Array
1 mark

Q3
Merging
1 mark

3/3 answered

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[100], b[100], c[200];
4     int n1, n2, i;
5     printf("Enter the elements of an array:");
6     scanf("%d", &n1);
7     printf("Enter the elements of first array: \n");
8     for(i=0; i<n1; i++) {
9         scanf("%d", &a[i]);
10    }
11    printf("Enter the size of second array: ");
12    scanf("%d", &n2);
13    printf("Enter the elements of second array: \n");
14    for(i=0; i<n2; i++) {
15        scanf("%d", &b[i]);
16    }
17    for(i=0; i<n1; i++) {
18        c[i]=a[i];
19    }
20    for(i=0; i<n2; i++) {
21        c[n1 + i]=b[i];
22    }
23    printf("Merged array: \n");
24    for(i=0; i<n1 + n2; i++) {
25        printf("%d", c[i]);
26    }
27 }
28 }
```

Run **Save** **Submit**

Input

```
3
1 2 3
4
4 5 6 7
```

Output

```
Array:
Enter the size of second array:
Enter the elements of second array:
Merged array:
1234567
```

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1stQuestion :

CO1-Output Prediction

QUESTIONS

Q1
Searching
1 mark

Q2
Array
1 mark

Q3
Merging
1 mark

3/3 answered

Q1 Searching1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```
1 #include<stdio.h>
2 int main(){
3     int a[100],n,i,target,count=0;
4     printf("enter the size of the array: ");
5     scanf("%d",&n);
6
7     for(i = 0;i<n;i++){
8         scanf("%d",&a[i]);
9     }
10    printf("enter the target:");
11    scanf("%d",&target);
12    for(i = 0;i<n;i++){
13        if(a[i]==target){
14            count++;
15        }
16    }
17    printf("count: %d",count);
18 }
```

Submitted

Input

3
12 12 4
12

Output

enter the size of the array: enter
the target:count: 2